

Aayush Sood

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EDUCATION

University of Waterloo

Honours Software Engineering, Co-op

Waterloo, ON

2026 - Present

PROJECTS

IWKids - Mobile App | *React Native, JavaScript, Expo, Git, OpenAI API* ([GitHub](#))

- Engineered a cross-platform React Native app integrating the OpenAI API to automate pediatric ER workflows
- Reduced simulated triage data collection time by 40%, saving triage nurses an estimated 1-2 hours per shift
- Developed a working prototype in under 48 hours. Placed 2nd out of 24 teams at the Atlantic Canada AI summit

SAP-1 - 6502 Microprocessor / 8-bit Project | *Python, Assembly* ([GitHub: 6502](#)) ([GitHub: 8-bit](#))

- Engineered and assembled functional 8-bit and 6502 breadboard computers using 74LS-series discrete logic chips and SAP-1 architecture to execute hardware-level CPU operations
- Programmed low-level routines in Assembly to run working programs on the physical hardware, debugged power, memory addressing, and signal issues directly on the breadboard

CoinDeno | *React.js, React Native, REST, Git, Firebase, CoinGecko APIs* ([GitHub](#)) ([Link](#))

- Grew an active user base to 20+ cross-platform users by launching a crypto tracker on the App Store and Google Play, built with React Native, CoinGecko and REST APIs
- Implemented secure Google OAuth authentication and integrated CoinGecko APIs for live portfolio tracking
- Accelerated feature deployment across iOS and Android (~85%) by engineering a unified reactive native architecture

Market Maker | *Python* ([GitHub](#))

- Engineered a Python-based trading system simulation that dynamically adjusts buy/sell spreads and asset pricing based on real-time market volatility and automated inventory-level logic
- Validated PnL & inventory models across 3 distinct trader profiles (Random, Momentum, Informed) using NumPy and Matplotlib to visualize automated execution metrics

EXPERIENCE

Stanford University's Code in Place - Section Lead

May 2025, May 2026

- Selected to lead weekly Python instruction for two cohorts totaling 30+ global learners as part of Stanford's flagship CS106A global education initiative
- Earned top-tier evaluation scores across 100% of weekly AI feedback reports by leading interactive code reviews, breakout discussions, and step-by-step Python instruction

Nova Scotia Secondary School Students' Association - IT Administrator

Apr 2025 - Jun 2026

- Maintained 100% ticket resolution across M365, Entra, and Google Workspace for 100+ provincial users by authoring documentation and providing end-to-end admin support
- Engineered a responsive redesign of www.nsssa.ca, supporting registration and content delivery for hundreds of annual conference delegates, by modernizing the page architecture, layout responsiveness, and asset loading

Tech Club President

Sep 2023 - Jun 2026

- Expanded student membership by 800% (from 3 to 30+ active members) by designing and facilitating 80+ technical workshops in web development, AI, cybersecurity, and embedded systems
- Executed 5 large-scale tech events averaging 40+ attendees and managing a ~\$5k budget by coordinating event logistics, securing funding, and leading club operations

HONORS & AWARDS

- Schulich Leader Scholarship (McGill):** Winner of Canada's premier \$100k STEM undergraduate scholarship (2026)
- Scotiabank Software Engineering Scholarship (UWaterloo):** \$20k entrance scholarship for outstanding students (2026)
- Wolf Scholar Finalist (UofT):** Named a national finalist for Inaugural Wolf Scholarship (1 of 32 selected nationwide) (2026)
- AFA CyberPatriot:** Won Gold Tier in an international, team-based cyber defense competition (2025)
- IWK AI Pitch Competition:** Won 2nd Place for designing an AI-driven pediatric healthcare workflow solution (2025)
- Lockheed Martin CYBERQUEST Competition:** Won 3rd Place in a team-based, global cybersecurity CTF challenge (2025)
- Canadian Computing Competition Senior Division:** Certificate of Distinction, Score: 45/75 (2025)
- Canadian Computing Competition Junior Division:** Certificate of Distinction & Honour Rolls, Score: 66/75 (2024)

Skills

Languages: Python, C++, JavaScript, SQL

Frameworks: React, React Native, Node.js, HTML, CSS/SCSS, Tailwind, REST APIs

Tools: Firebase, Cloudflare, Netlify, Git, GitHub, MS Azure, Linux, Arduino, Claude Code, OpenAI API